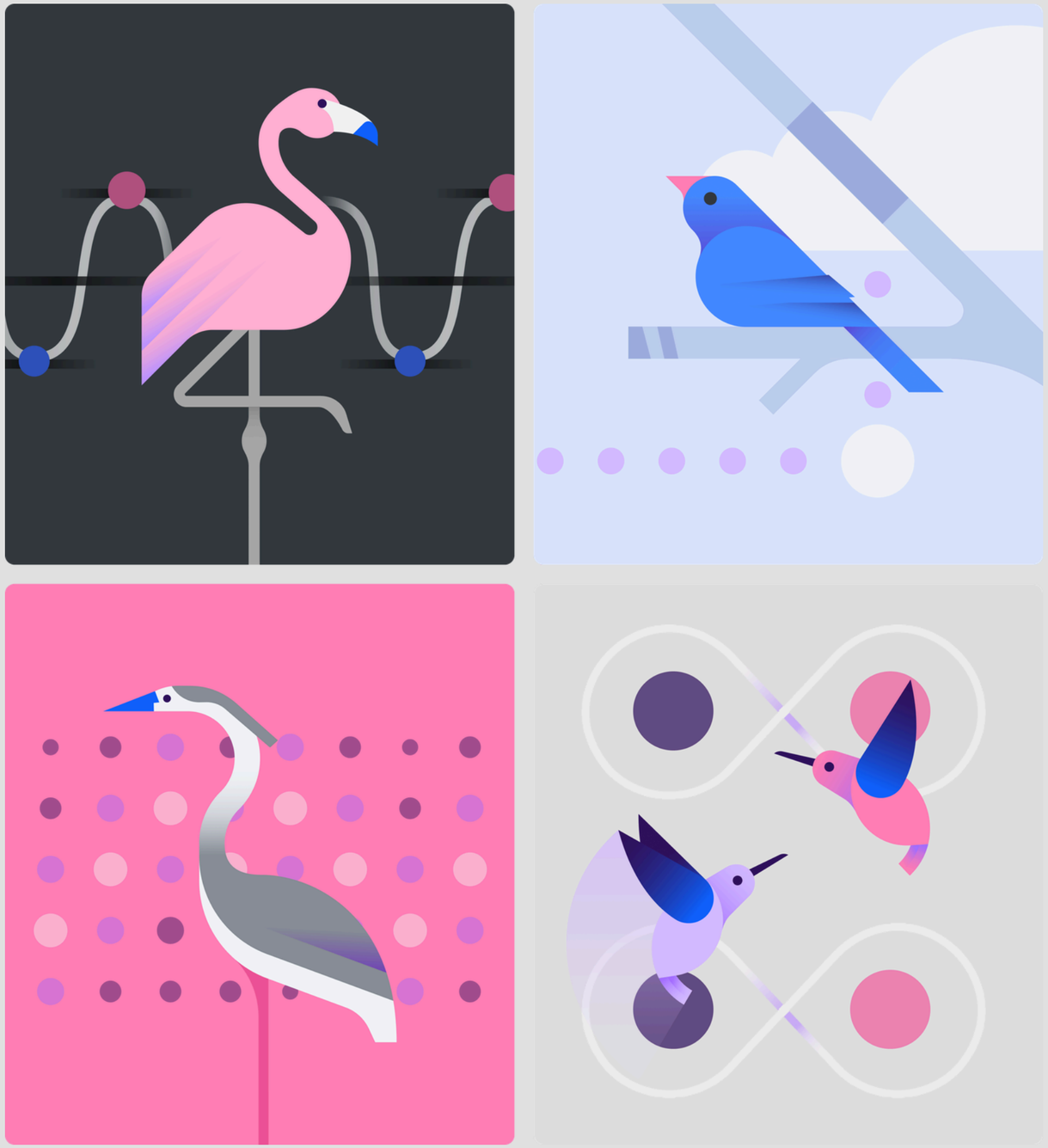
Qiskit Fall Fest 2026

QUANTUM FOURIER TRANSFORM (QFT)

Aranya Mukherjee
4th Year BS-MS
Physics Major





Fourier Transforms

What & Why?

Where It All Started!

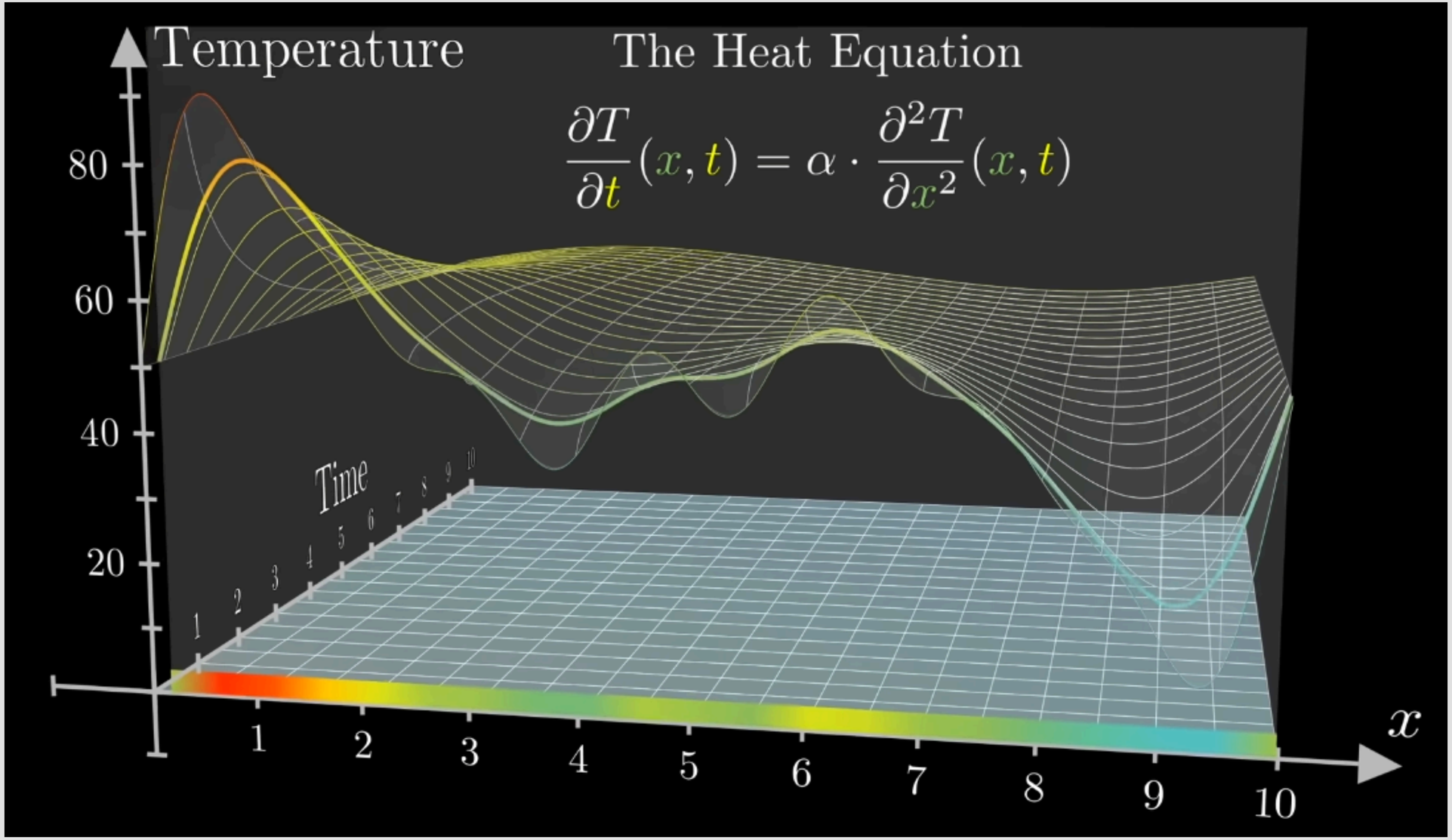
The Heat Equation

$$\frac{\partial T(x,t)}{\partial t} = \alpha \frac{\partial^2 T(x,t)}{\partial x^2}$$



Joseph Fourier

Where It All Started!



Where it all started: The Heat Equation

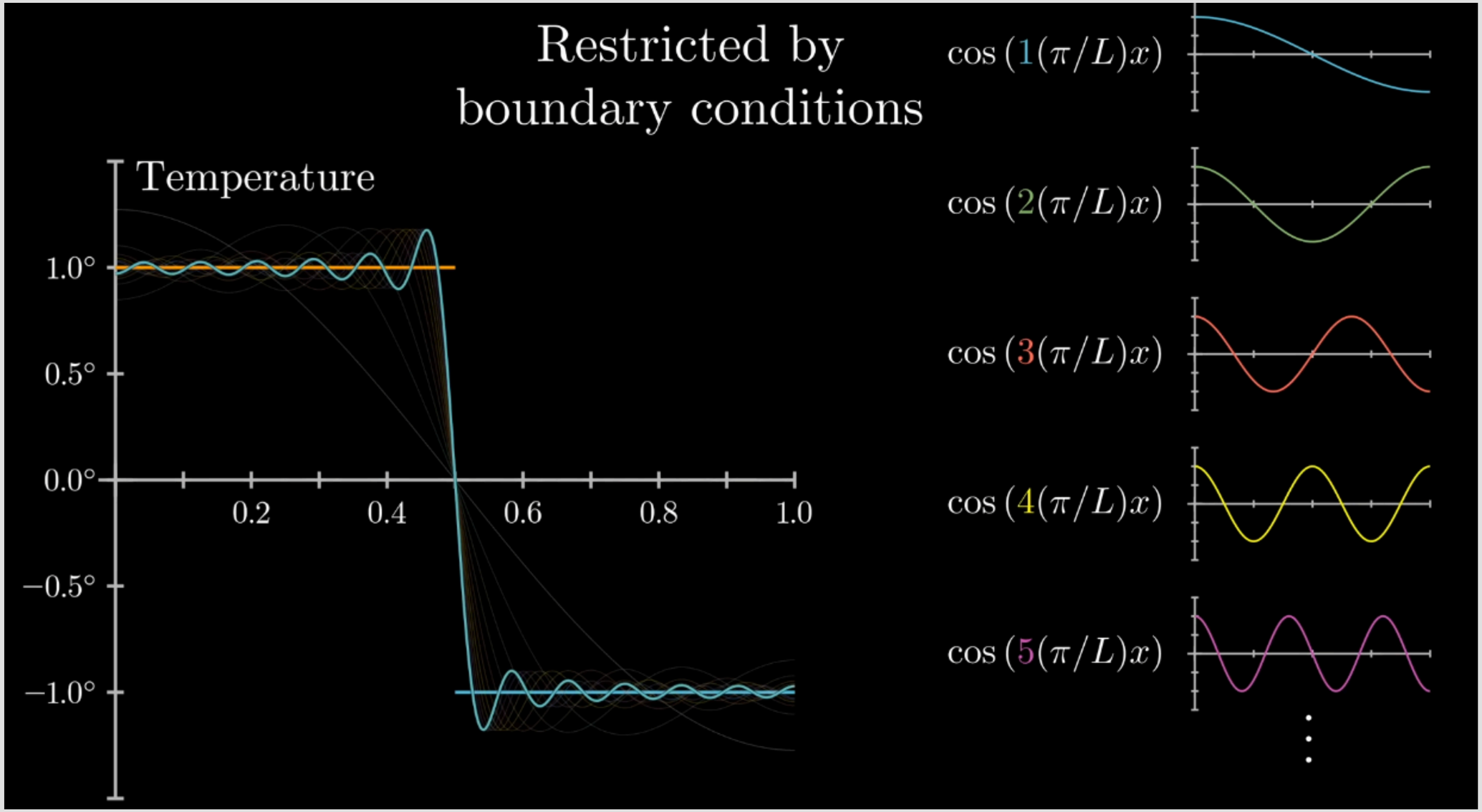
$$T(x, 0) = \sin(kx)$$



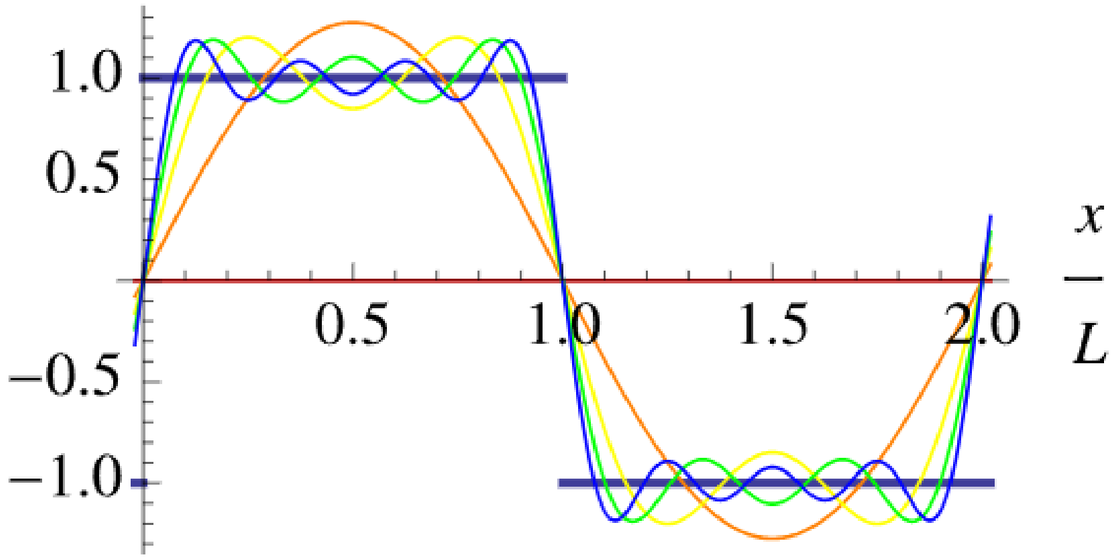
$$T(x, t) = e^{-\alpha k^2 t} \sin(kx)$$

Decompose any signal into sines and cosines

Where it all started: The Heat Equation

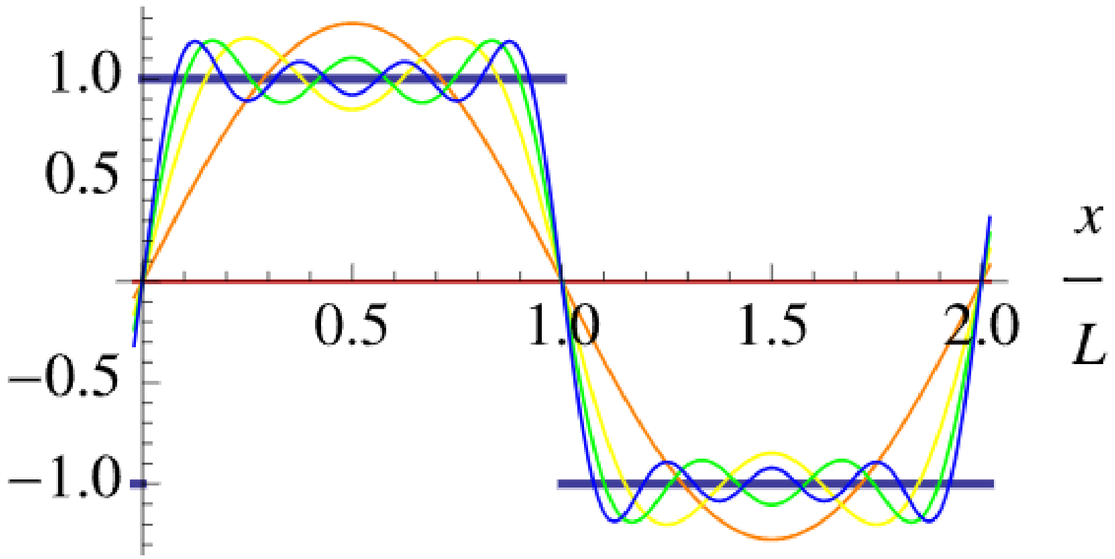


Where it all started: The Heat Equation



$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

Where it all started: The Heat Equation



$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$e^{ix} = \cos x + i \sin x$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx$$

Fourier Transforms



Fourier Transform → Continuous extension of Fourier series for non-periodic functions as signal's period approaches infinity

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

Fourier Transforms

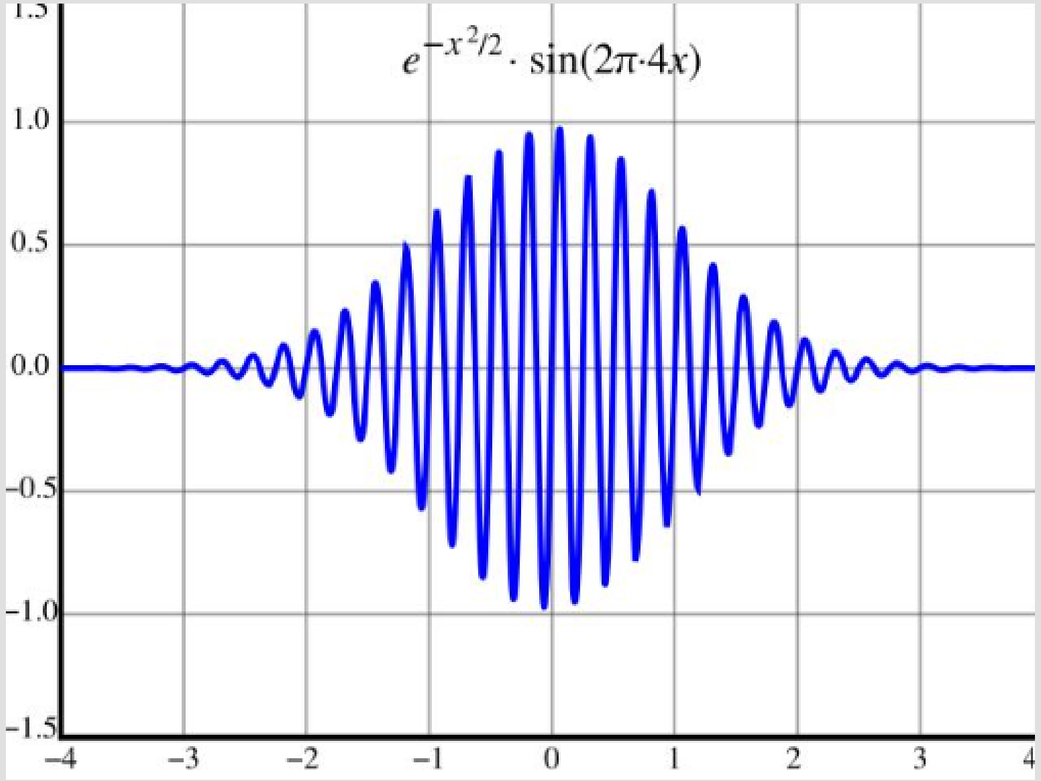


Fourier Transform → Continuous extension of Fourier series for non-periodic functions as signal's period approaches infinity

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

Let's decode the math!

Fourier to Uncertainty Relations



- A short-lived signal generally needs a broad range of frequencies
- A nearly pure tone can persist for a long time

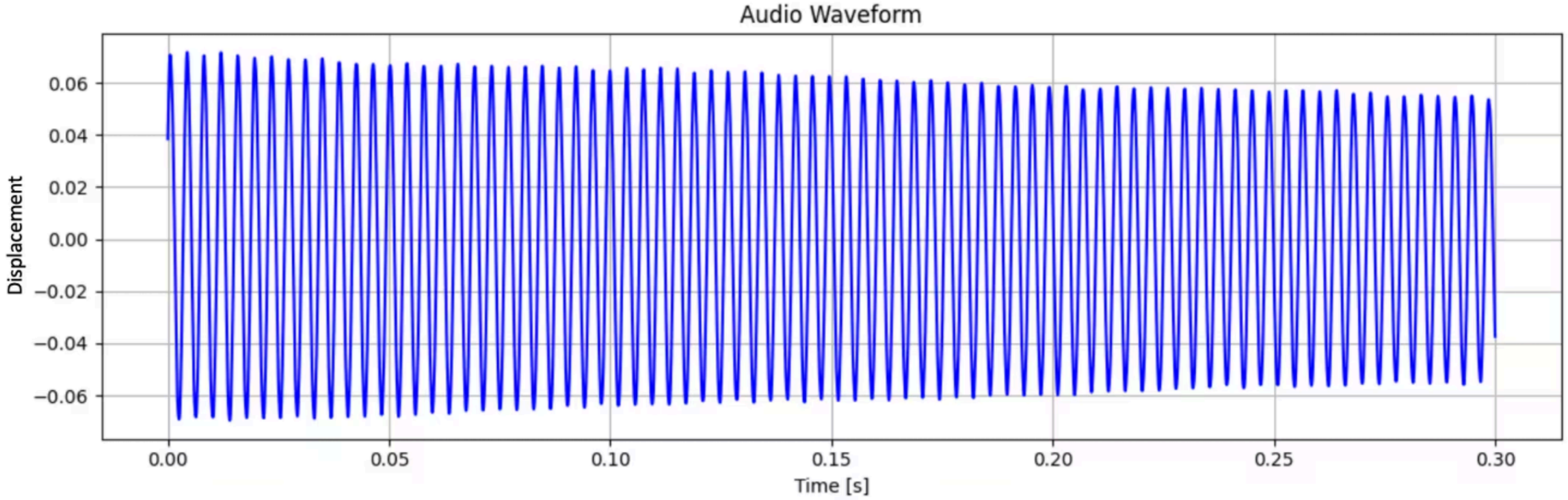
$$\Delta x \Delta p \geq \frac{\hbar}{4\pi}$$

Shorter in Time → Broader in frequency

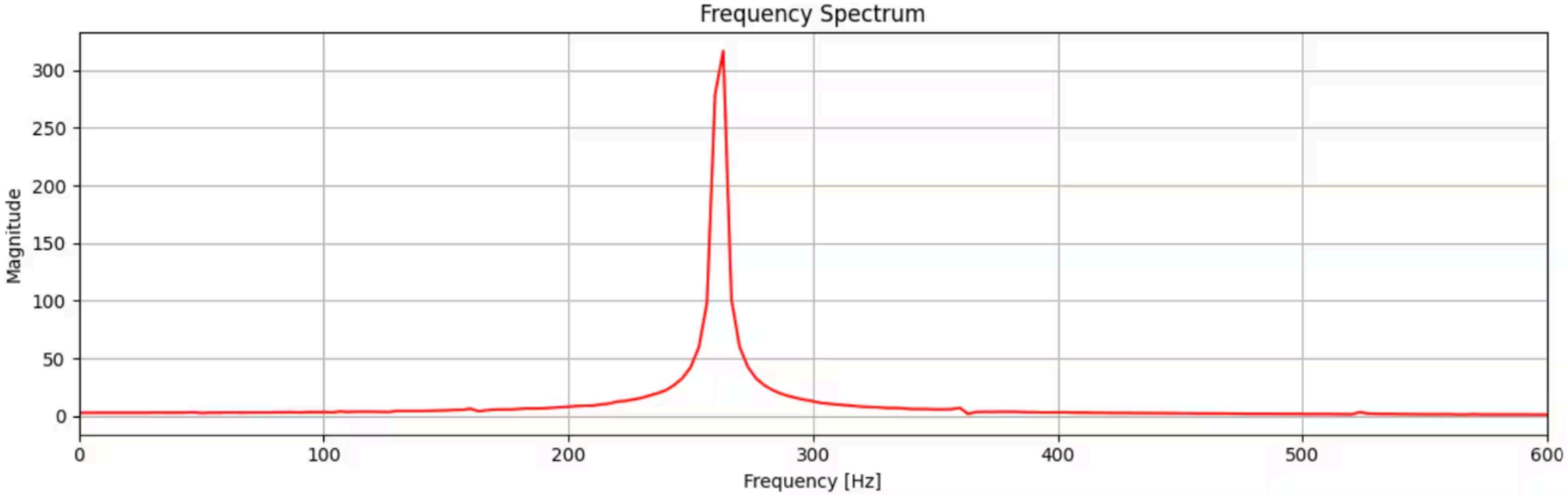
Fourier Transforms (with Music!)



Let’s play the Middle C!

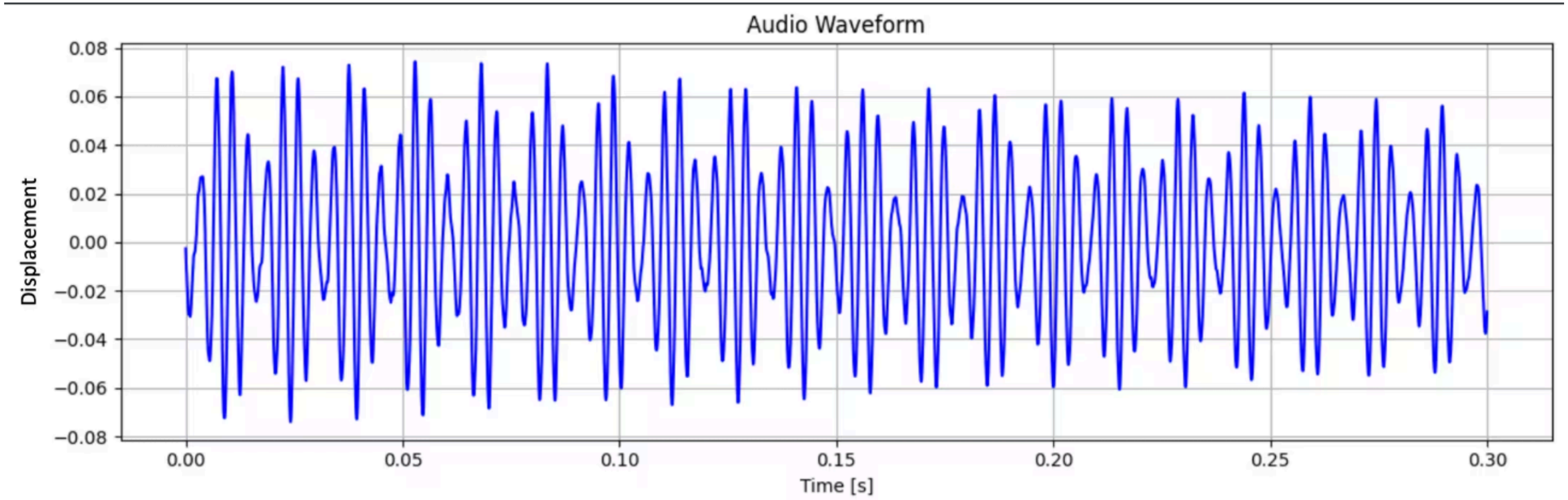


Fourier Transforms (with Music!)

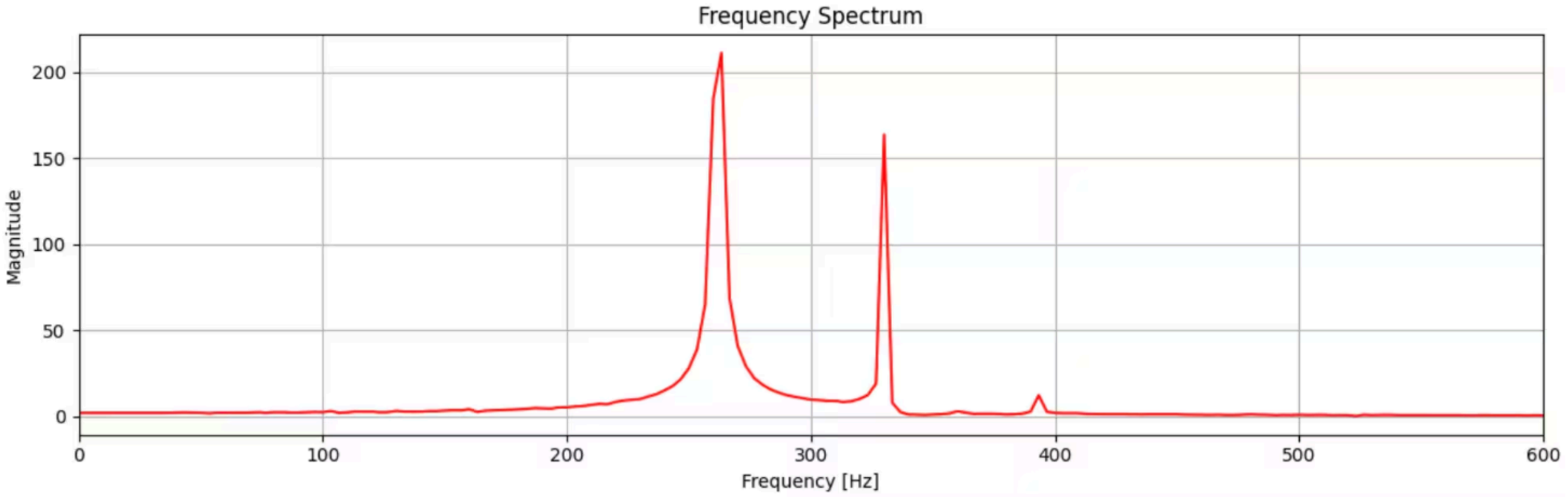


Fourier Transforms (with Music!)

Now, let’s play the C-Major Chord(Notes C,E,G)

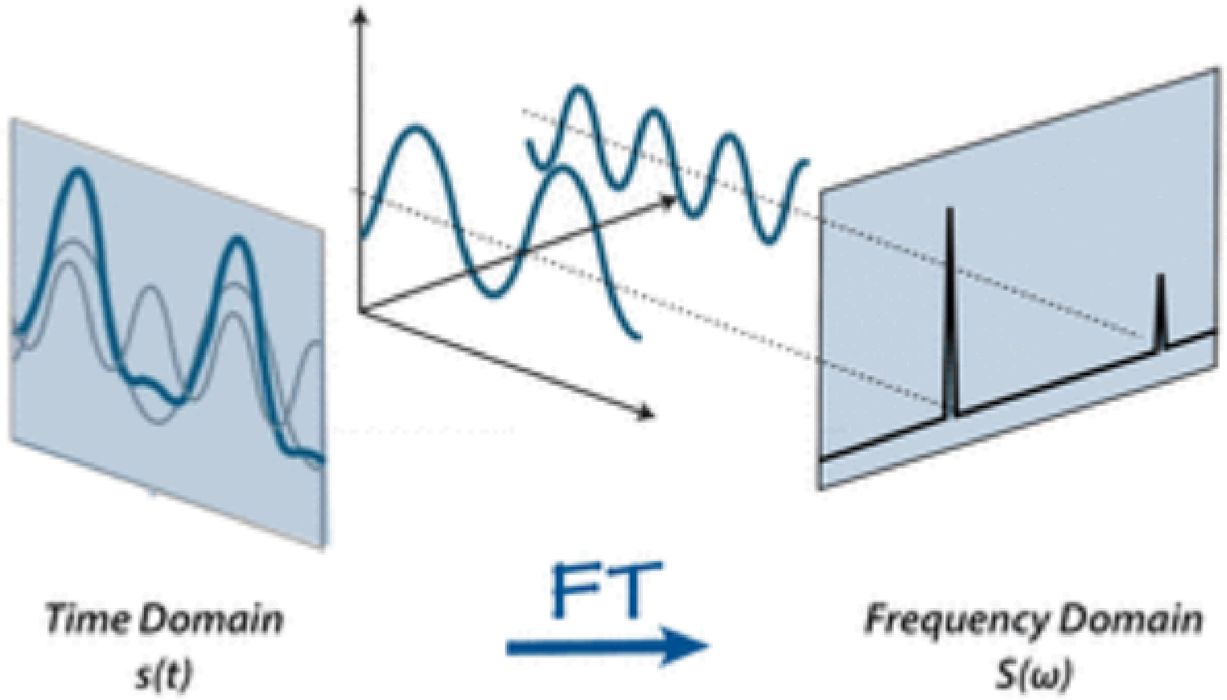


Fourier Transforms (with Music!)



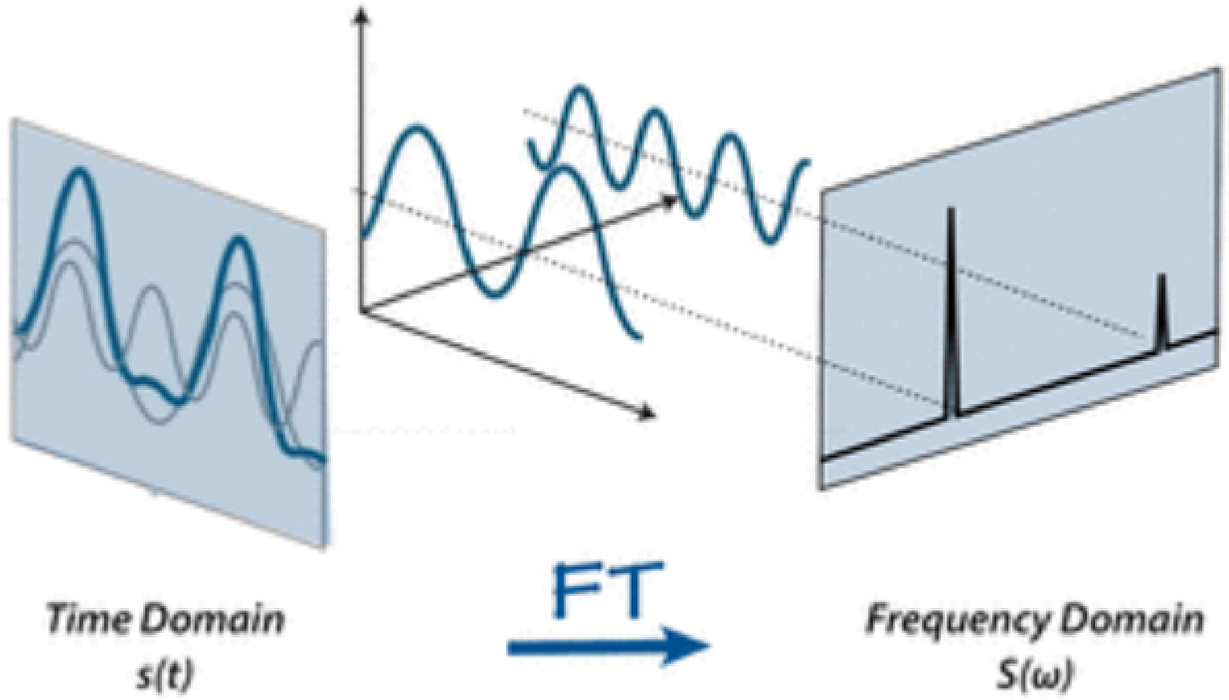
Introducing FTM

Fourier Transform Machine



Introducing FTM

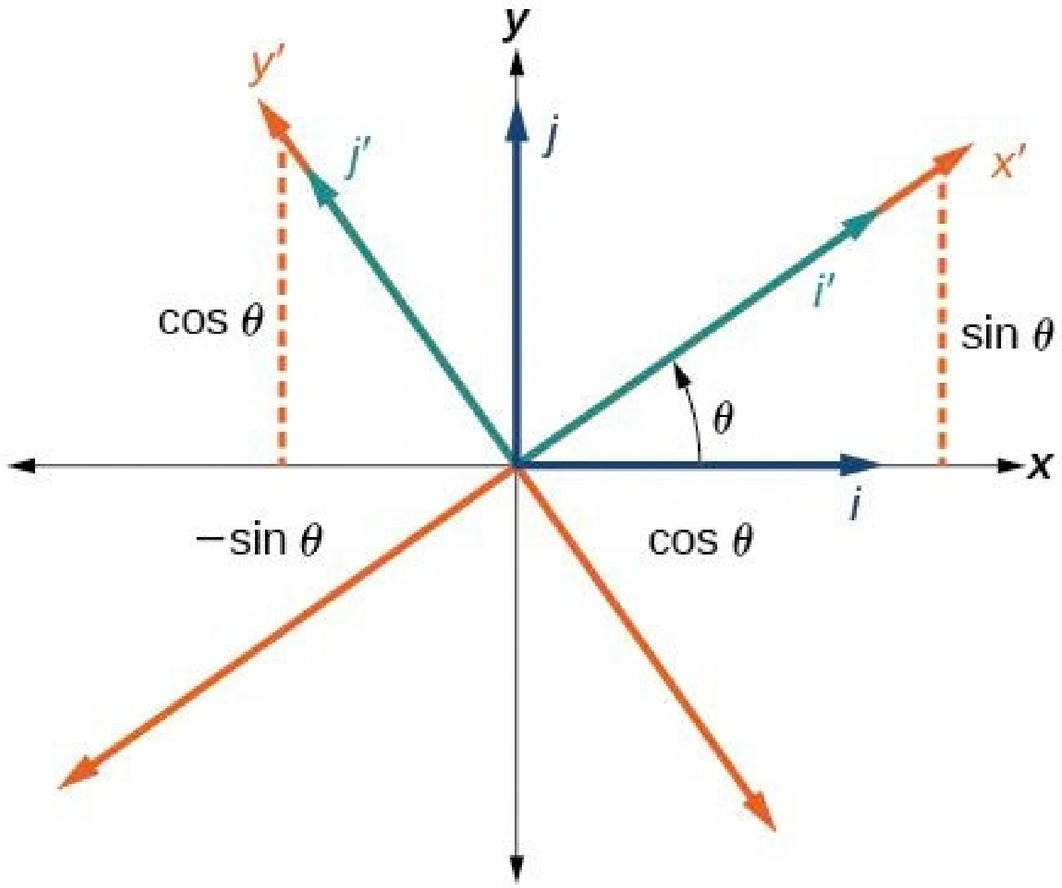
Fourier Transform Machine



- The underlying physics doesn’t change
- Just changing basis in which signal is described

Sound Familiar?

Introducing FTM

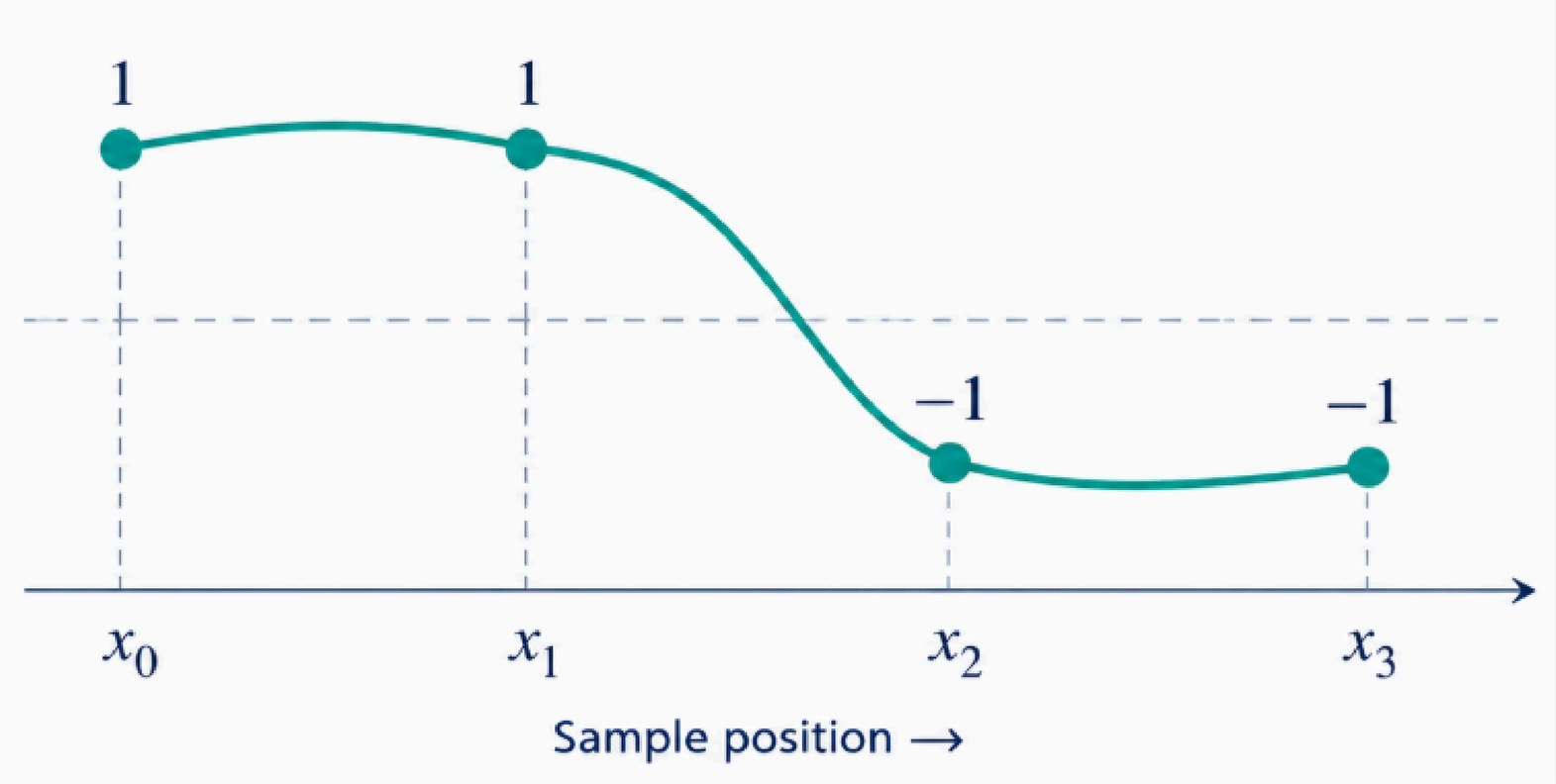


$$\vec{v} = v_x\hat{x} + v_y\hat{y}$$

$$\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$

$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Discrete Fourier Transform(DFT)



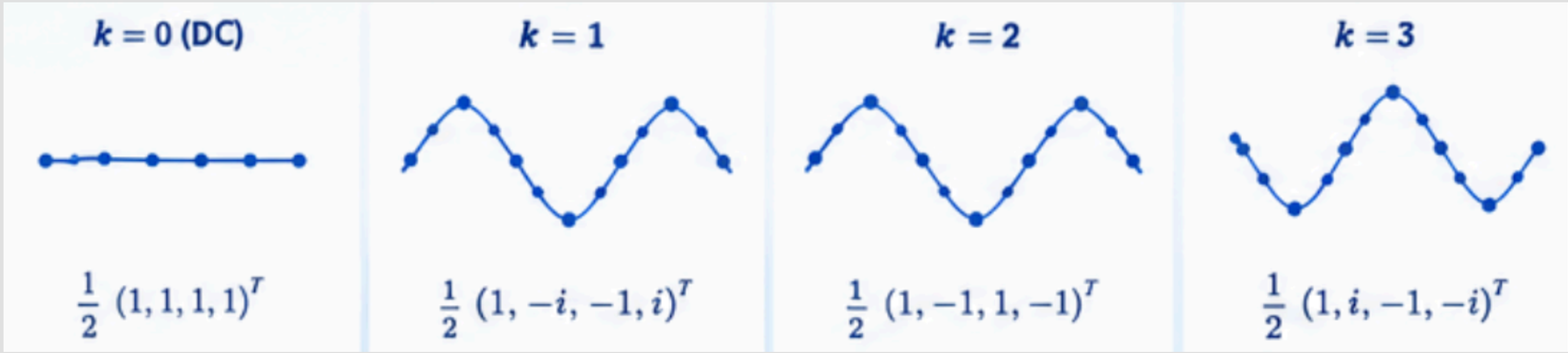
$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

$$\mathbf{X} = F_4\mathbf{x}$$

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1-i \\ 0 \\ 1+i \end{pmatrix}$$

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{2\pi i j k/N}$$

Discrete Fourier Transform(DFT)



In sample space:

- Vector → Point displacements
- Difficult to see underlying pattern
- All components needed

In Fourier space:

- Vector → Frequency Amplitudes
- Reveals oscillatory patterns(simpler)

Let’s put the Quantum in Quantum Fourier Transform(QFT)



Entering the Quantum World

$$|\psi\rangle = \sum_{j=0}^{N-1} a_j |j\rangle$$

$|j\rangle$: Computational Basis states

a_j : Complex probability amplitudes

$$|j\rangle = |00\ldots00\rangle, |00\ldots01\rangle, |00\ldots11\rangle, \ldots |11\ldots11\rangle$$

$$|11\ldots11\rangle = |2^m - 1\rangle = |N-1\rangle$$

Entering the Quantum World

$$\mathrm{QFT}|j\rangle = |f_j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k/N} |k\rangle$$

N : number of basis states
m : Number of qubits
$N = 2^m$

$$\mathrm{QFT}|j\rangle = |f_j\rangle$$

$$\mathrm{QFT}^\dagger |f_j\rangle = |j\rangle$$

Properties:

$$\mathrm{QFT}^\dagger \mathrm{QFT} = I$$

$$|\langle k|f_j\rangle|^2 = \frac{1}{N}$$

Building the intuition

Consider the qubit system in the basis:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Phase difference of pi not reflected in measurement

P(0) = 1/2, P(1) = 1/2

$$\mathrm{QFT}|+\rangle = |0\rangle \qquad \mathrm{QFT}|-\rangle = |1\rangle$$

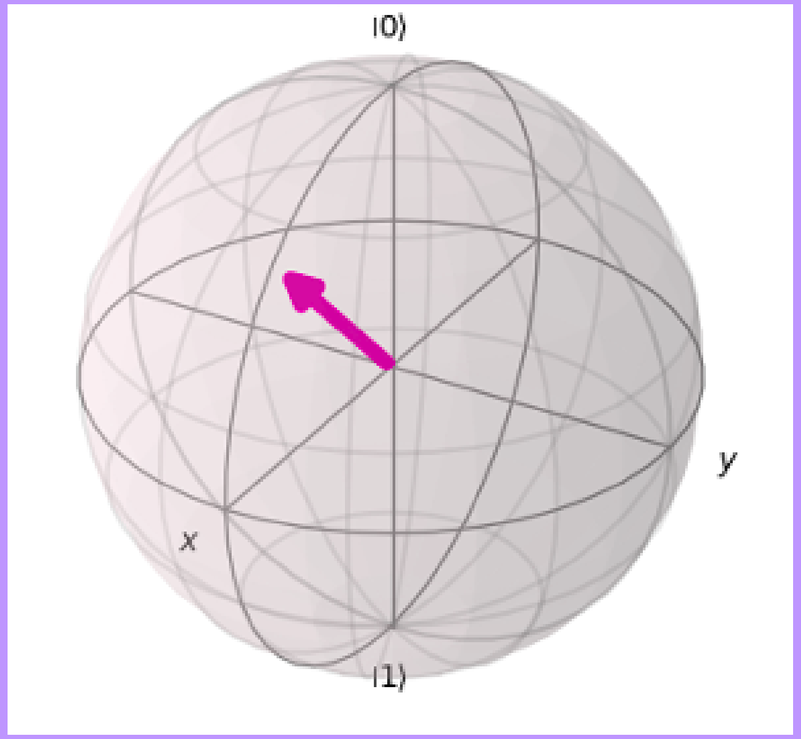
Building the intuition

Consider the qubit system in the basis:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\mathrm{QFT}|+\rangle = |0\rangle \quad \mathrm{QFT}|-\rangle = |1\rangle$$

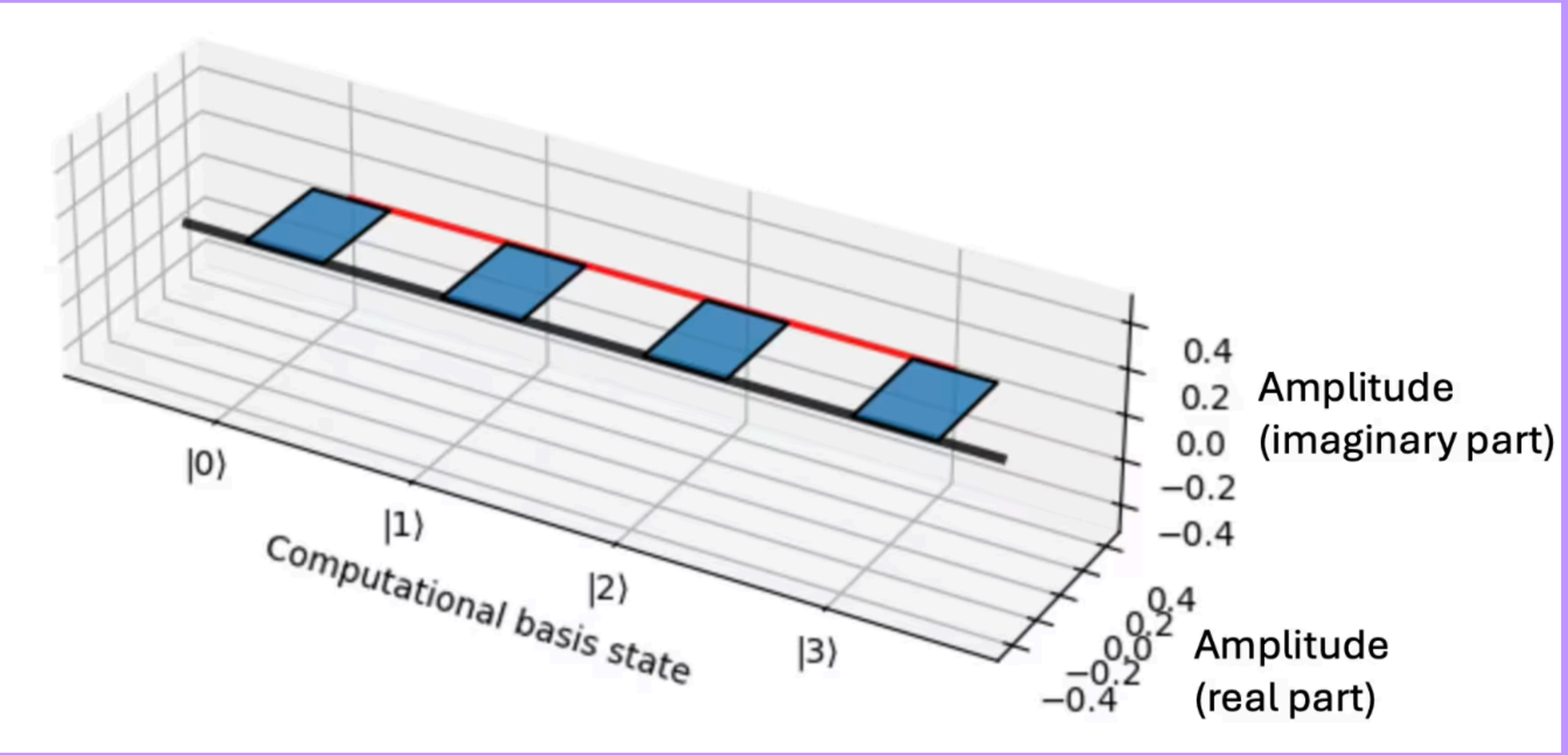
$$\mathrm{QFT}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H$$



Quantum Fourier States

m=2, N= 4

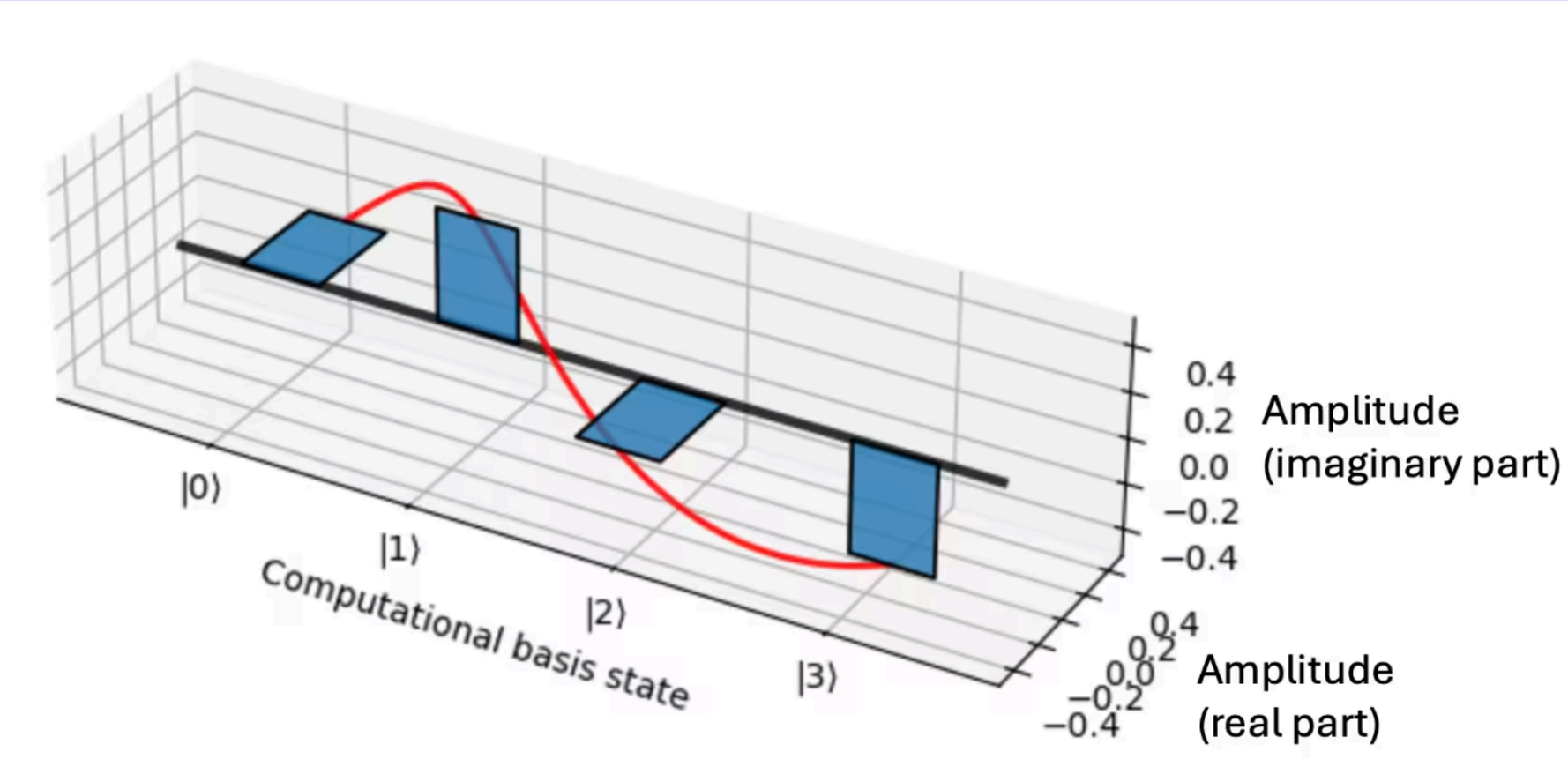
$$|\phi_0\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$



Quantum Fourier States

m=2, N= 4

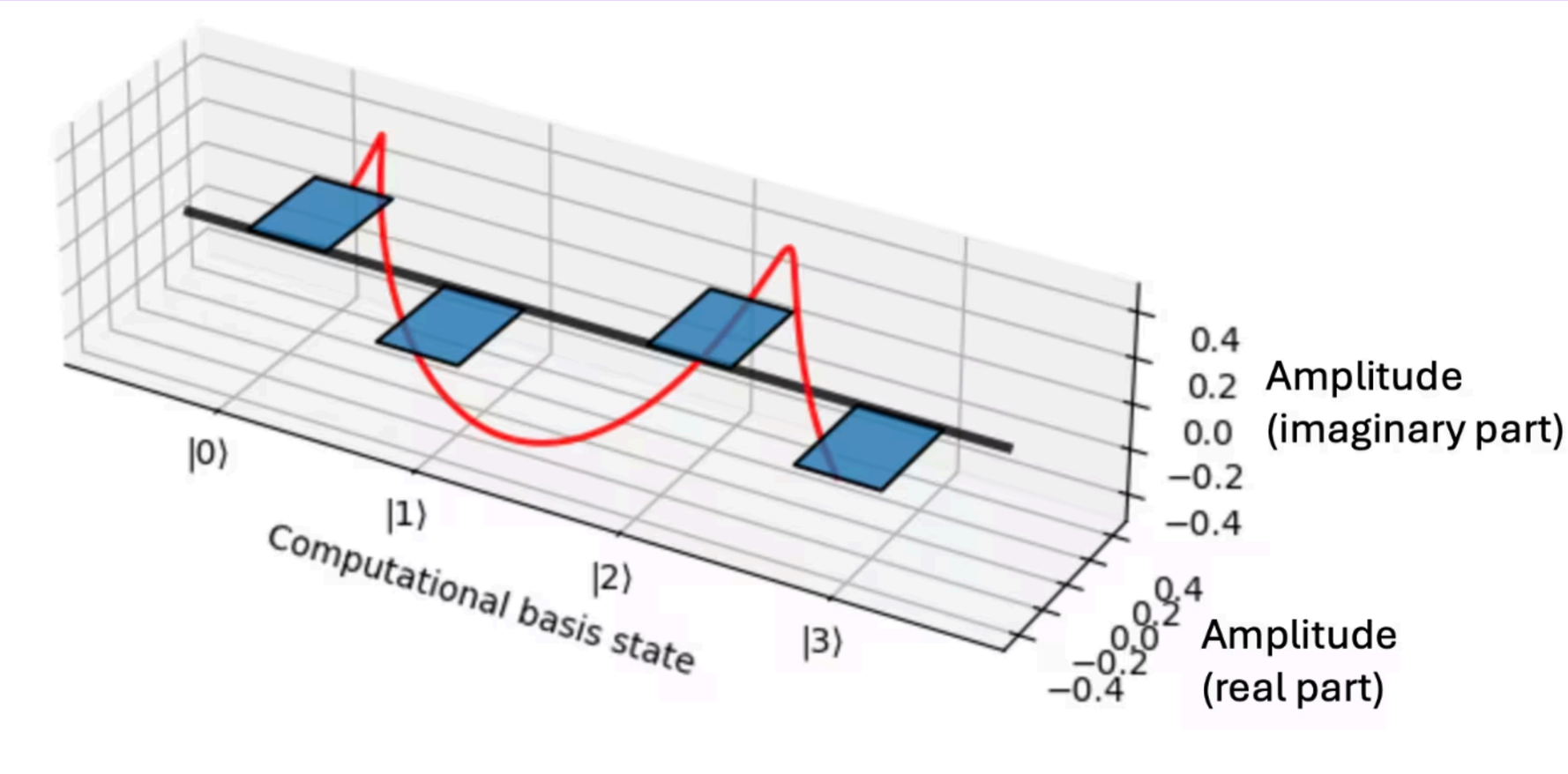
$$|\phi_1\rangle = \frac{1}{2}(|00\rangle + e^{i\pi/2}|01\rangle + e^{i\pi}|10\rangle + e^{i3\pi/2}|11\rangle)$$



Quantum Fourier States

m=2, N= 4

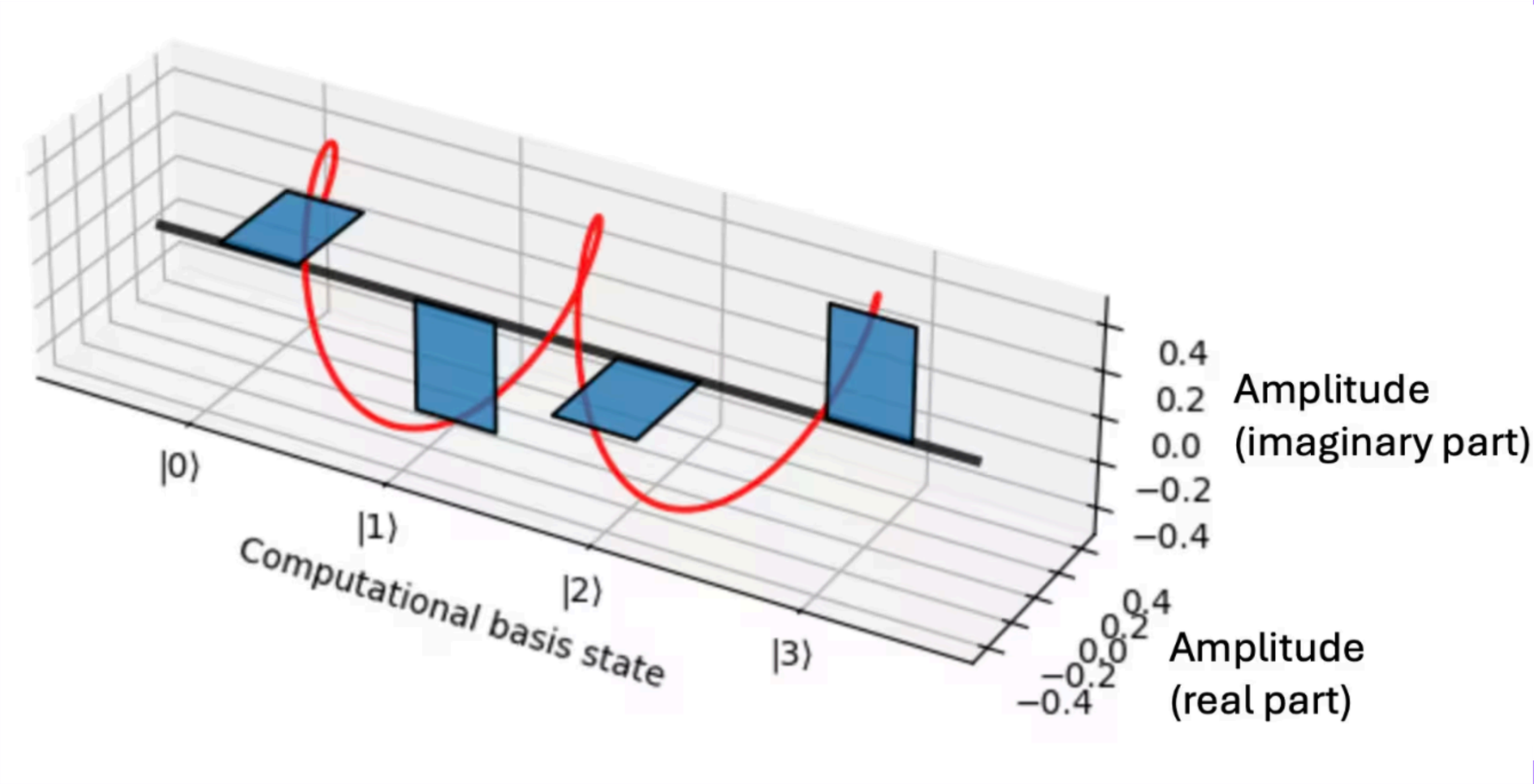
$$|\phi_2\rangle = \frac{1}{2}(|00\rangle + e^{i\pi}|01\rangle + e^{i2\pi}|10\rangle + e^{i3\pi}|11\rangle)$$



Quantum Fourier States

m=2, N= 4

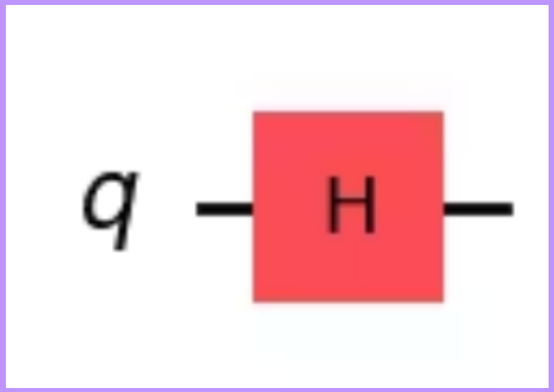
$$|\phi_3\rangle = \frac{1}{2}(|00\rangle + e^{i3\pi/2}|01\rangle + e^{i6\pi/2}|10\rangle + e^{i9\pi/2}|11\rangle)$$



QFT: The Algorithm

For single-Qubit system, (N=2)

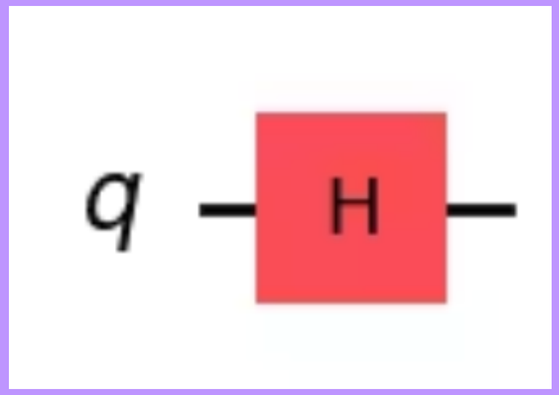
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QFT: The Algorithm

For single-Qubit system, (N=2)

$$\mathrm{QFT}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = H$$



For two qubits (N=4)

$$\mathrm{QFT}_4 = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & i & -1 & -i\\ 1 & -1 & 1 & -1\\ 1 & -i & -1 & i\end{pmatrix}$$

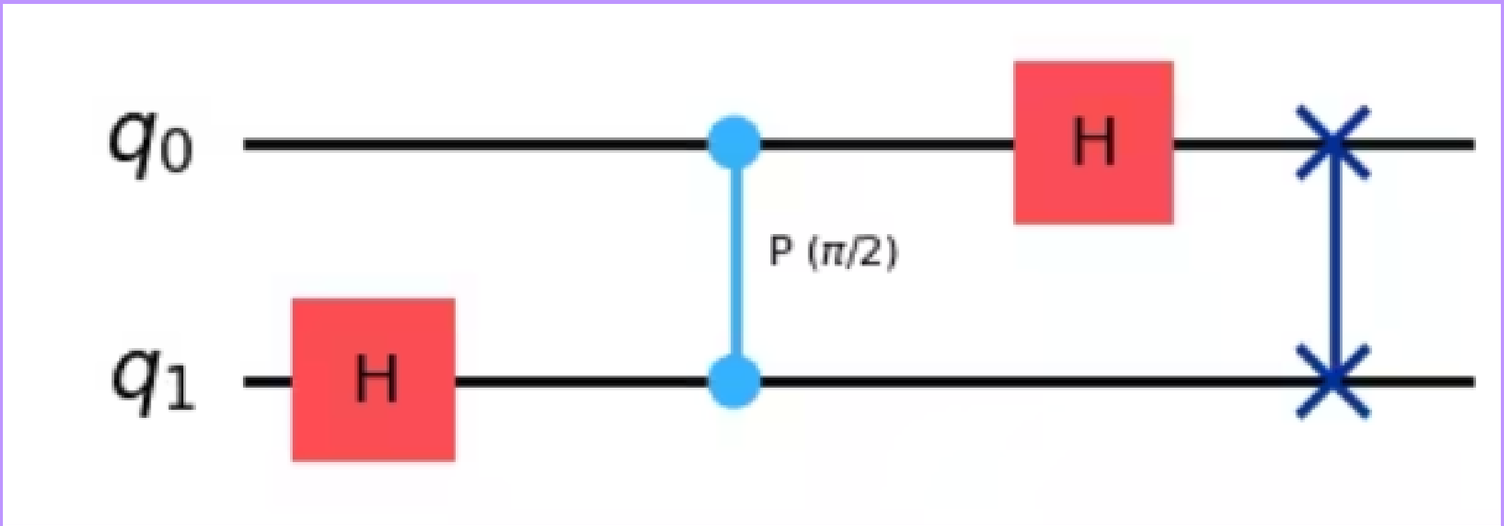
QFT: The Algorithm

For two qubits (N=4)

$$\mathrm{QFT}_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

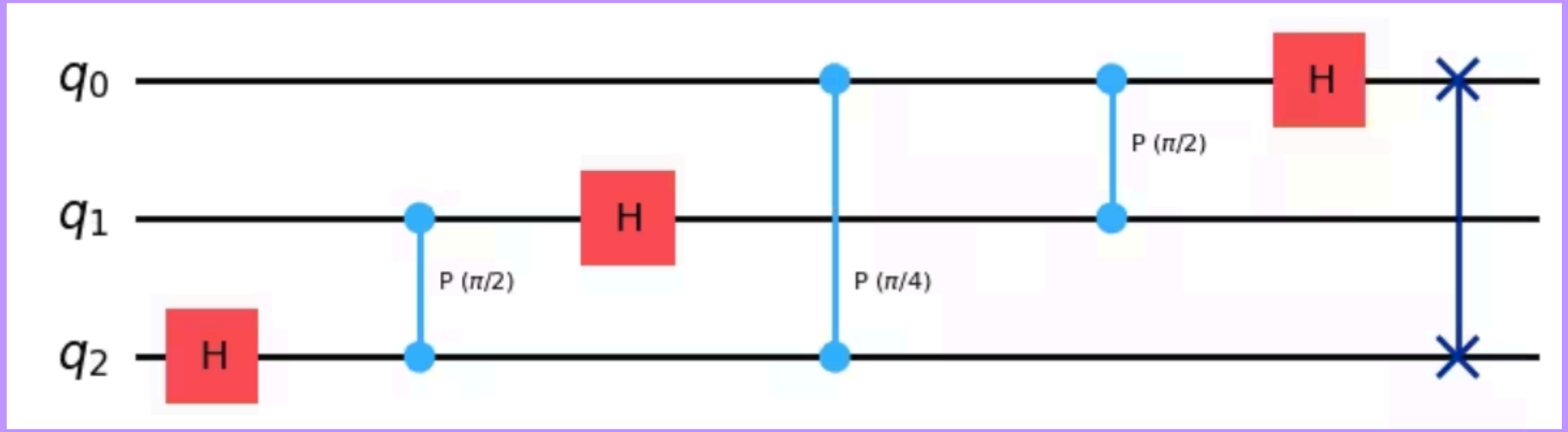
$$\mathrm{CP}_\alpha = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix}$$

$$\mathrm{SWAP}_\alpha = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



QFT: The Algorithm

For three qubits (N=8)



And so on...

Applying QFT : QPE

Quantum Phase Estimation

Unitary Operators:

$$U|u\rangle = e^{2\pi i\theta}|u\rangle$$

θ is a real number between 0 and 1

We will see how QFT is used to determine these eigenvalues (phases imparted by unitary operators)

Applying QFT : QPE

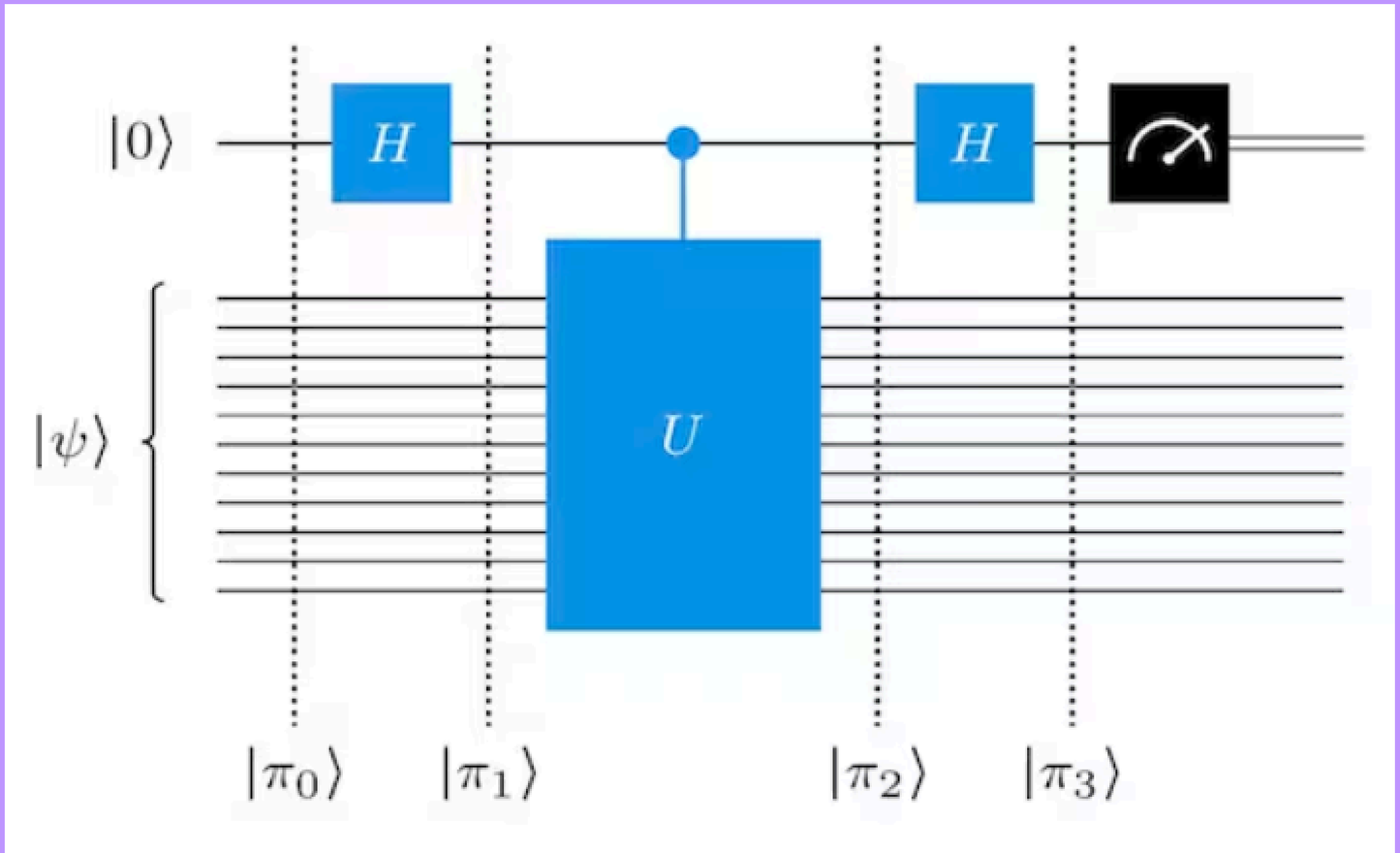
Algorithm

Qubit is prepared in the state $|\pi_0\rangle = |\psi\rangle|0\rangle$

$\text{where} |\psi\rangle \text{is an eigenstate of unitary operator U}$

Applying QFT : QPE

Algorithm



$$|\pi_0\rangle = |\psi\rangle|0\rangle$$

$$|\pi_1\rangle = \frac{1}{\sqrt{2}}|\psi\rangle(|0\rangle + |1\rangle)$$

$$|\pi_2\rangle = \frac{1}{\sqrt{2}}(|\psi\rangle|0\rangle + e^{2\pi i\theta}|\psi\rangle|1\rangle)$$

$$|\pi_3\rangle = \frac{1}{\sqrt{2}}|\psi\rangle(\frac{1+e^{2\pi i\theta}}{2}|0\rangle + \frac{1-e^{2\pi i\theta}}{2}|1\rangle)$$

$$= |\psi\rangle(\cos(\pi\theta)|0\rangle - i\sin(\pi\theta)|1\rangle)$$

Takeaways



- **Fourier transforms decompose complex signals into simpler frequency components revealing hidden structures.**
- **The QFT is a quantum operation, not merely a change of notation**. It transforms quantum amplitudes, allowing relative phases to interfere and influence measurement outcomes.
- **Quantum phase carries useful information**. Although phase cannot generally be read directly from a single measurement, quantum interference can convert phase information into measurable probabilities.



Thank You!

On to some Quantum Computing!



